# Introduction to Data Analytics
# MS - 4160

Session 10 – KNN and concepts

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# Supervised Learning: Concepts

- Data versus Algorithmic modelling: KNN
- Bias-Variance Dichotomy
- Regularization and coefficient shrinkage
- Model Validation
- Dimensionality reduction

# The difference of the K-Nearest Neighbours approach

- Breiman, L. (2001). Statistical modeling: The two cultures (with comments and a rejoinder by the author). *Statistical Science*, *16*(3), 199-231.

- In Statistical Learning:
  - Data modelling

$$y(x_1, x_2, \ldots, x_n) = \beta_0 + \sum_{i=1}^{n} \beta_i x_i + \varepsilon$$

    - e.g., Multiple regressions, Discriminant Analysis, etc.,

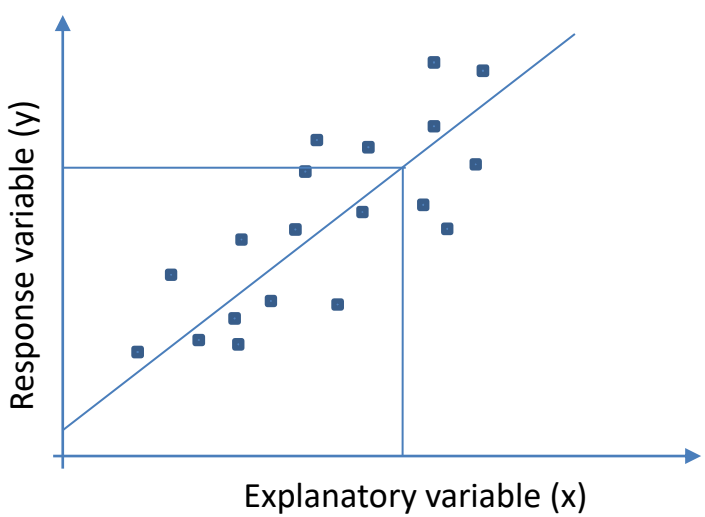
  - Algorithmic modelling
    - A set of algorithmic instructions that relate the independent and dependent variables
    - e.g., K-Nearest Neighbours, Random forests, etc.,

# KNN-Introduction

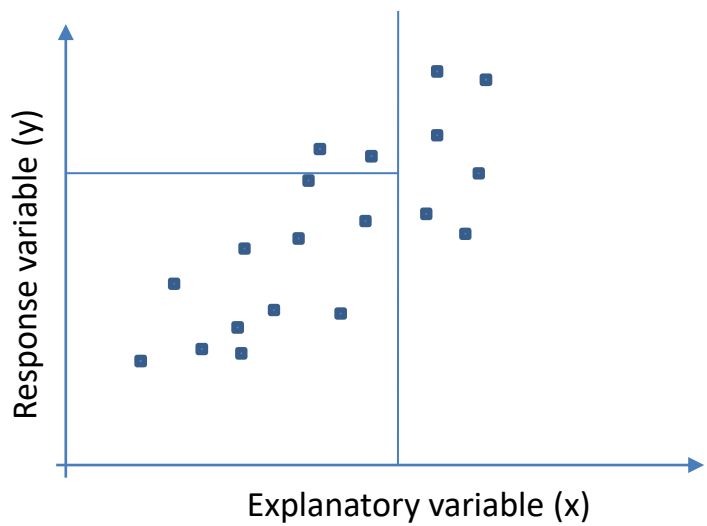
- The kNN classification is performed using the following four steps:
- Compute the distance metric between the test datapoint and all the labeled data points
- Order the labeled data points in the increasing order of this distance metric
- Select the top k labeled data points and look at the outptut
- Decide output based on majority or average

# Prediction

The Regression Approach

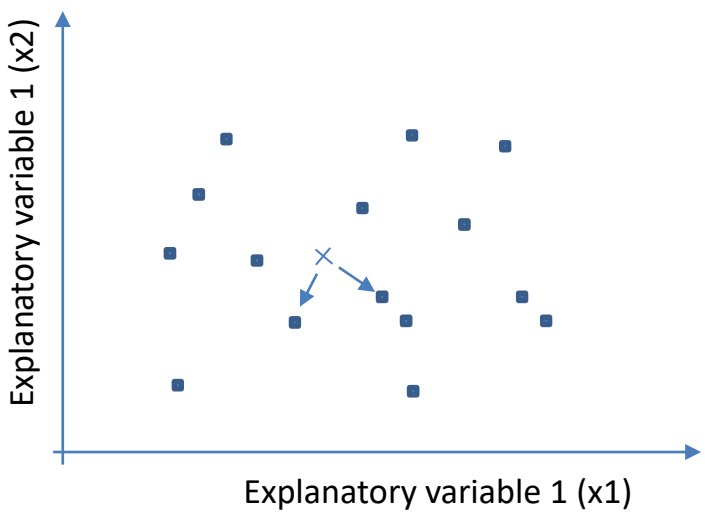


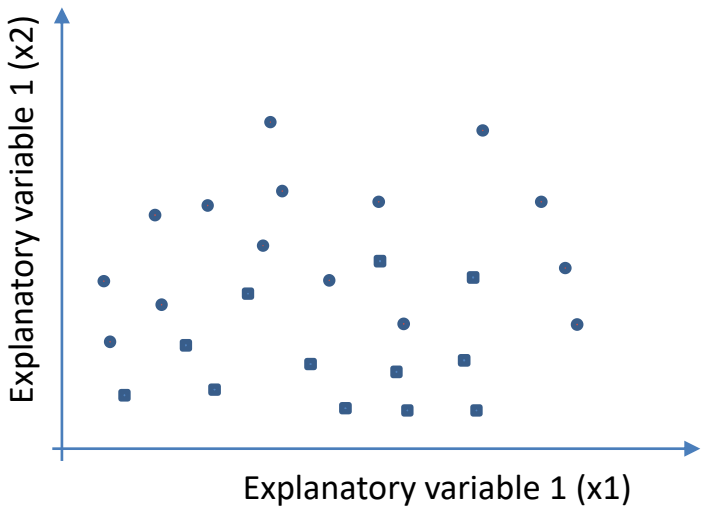

The K-NN approach



# Prediction

- KNN description for two inputs as well as classification problems



# K-NN

- Computational difficulties of a K-NN style approach
- What assumptions of the mathematical model is being made?
- What assumptions of the decision boundaries?
- What is the effect of outliers?
- What is the effect of distant points?
- What are some variants?

# Supervised Learning: Concepts

- Data Modelling versus Algorithmic modelling
- Bias-Variance Dichotomy
- Regularization and coefficient shrinkage
- Model Validation
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# The Math

$$
\begin{aligned}
\mathbb{E}[(f(x)-\hat{f}(x))^2] &= \mathbb{E}\left[\left((f(x)-\mathbb{E}[\hat{f}(x)])-(\hat{f}(x)-\mathbb{E}[\hat{f}(x)])\right)^2\right] && (4)\\
&= \mathbb{E}\left[\left(\mathbb{E}[\hat{f}(x)]-f(x)\right)^2\right]+\mathbb{E}\left[\left(\hat{f}(x)-\mathbb{E}[\hat{f}(x)]\right)^2\right]\\
&\quad -2\mathbb{E}\left[\left(f(x)-\mathbb{E}[\hat{f}(x)]\right)\left(\hat{f}(x)-\mathbb{E}[\hat{f}(x)]\right)\right] && (5)\\
&= \underbrace{(\mathbb{E}[\hat{f}(x)]-f(x))}_{=\mathrm{bias}[\hat{f}(x)]}{}^2+\underbrace{\mathbb{E}\left[\left(\hat{f}(x)-\mathbb{E}[\hat{f}(x)]\right)^2\right]}_{=\mathrm{var}(\hat{f}(x))}\\
&\quad -2\left(f(x)-\mathbb{E}[\hat{f}(x)]\right)\mathbb{E}\left[\left(\hat{f}(x)-\mathbb{E}[\hat{f}(x)]\right)\right] && (6)\\
&= \mathrm{bias}[\hat{f}(x)]^2+\mathrm{var}(\hat{f}(x))\\
&\quad -2\left(f(x)-\mathbb{E}[\hat{f}(x)]\right)\left(\mathbb{E}[\hat{f}(x)]-\mathbb{E}[\hat{f}(x)]\right) && (7)\\
&= \mathrm{bias}[\hat{f}(x)]^2+\mathrm{var}(\hat{f}(x)) && (8)
\end{aligned}
$$

- https://towardsdatascience.com/the-bias-variance-tradeoff-8818f41e39e9#:~:text=Proof%20of%20bias%2Dvariance%20decomposition&text=(1)%20is

# Concept

- The concept here is that while adding complexity to the model might improve the fit, it need not improve the predictive accuracy on new data.
- It is also referred to as Occam's Razor

Model 1: $y = b_0 + b_1 x$

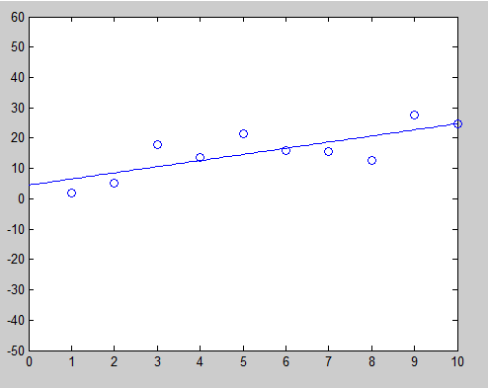
| x | y |
|---|---|
| 1 | 1.7 |
| 2 | 5.3 |
| 3 | 18.0 |
| 4 | 13.8 |
| 5 | 21.4 |
| 6 | 15.9 |
| 7 | 15.5 |
| 8 | 12.7 |
| 9 | 27.5 |
| 10 | 24.6 |

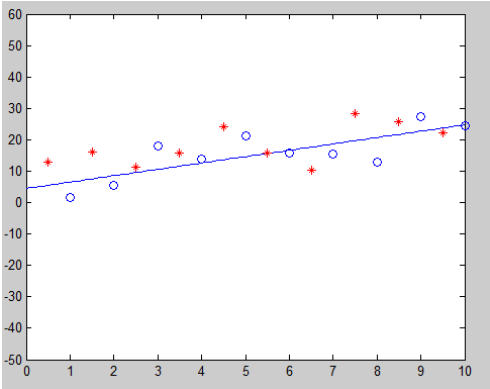
Model 2: $y = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + \cdots + b_9 x^9$

| x | x^2 | | x^9 | y |
|---|---|---|---|---|
| 1 | 1 | ... | 1 | 1.7 |
| 2 | 4 | | 262144 | 5.3 |
| 3 | 9 | | 4E+08 | 18.0 |
| 4 | 16 | | 7E+10 | 13.8 |
| 5 | 25 | | 4E+12 | 21.4 |
| 6 | 36 | | 1E+14 | 15.9 |
| 7 | 49 | | 2E+15 | 15.5 |
| 8 | 64 | | 2E+16 | 12.7 |
| 9 | 81 | | 2E+17 | 27.5 |
| 10 | 100 | | 1E+18 | 24.6 |

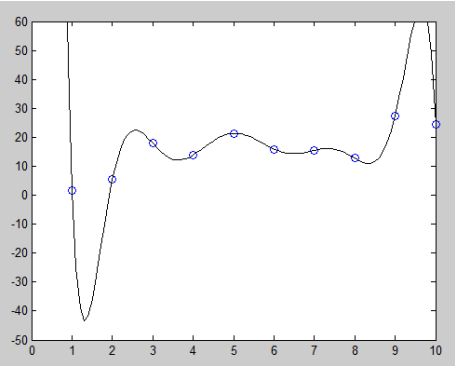
# Illustration

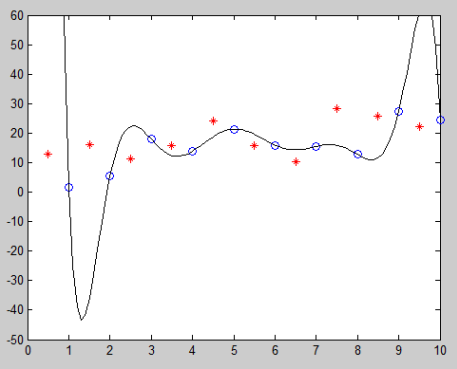
Linear Model




9th Order Polynomial Model

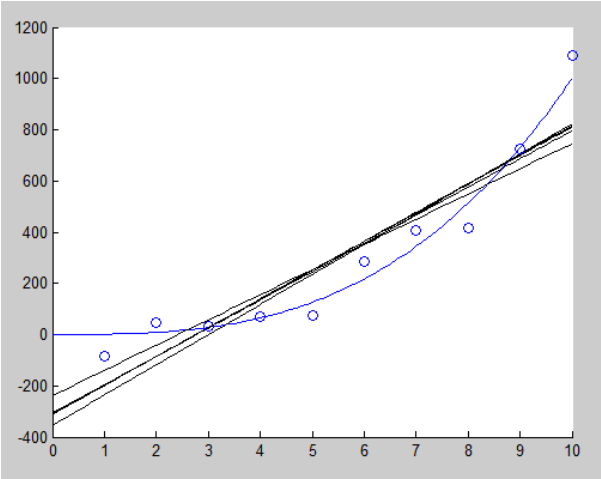


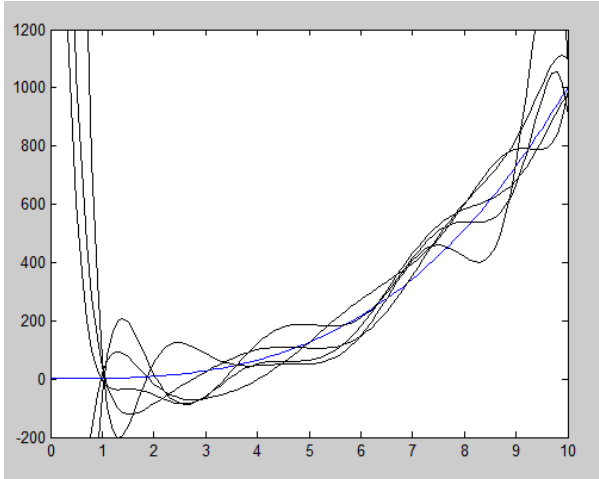
# Illustration 2

- What happens when there is more complexity in the system?

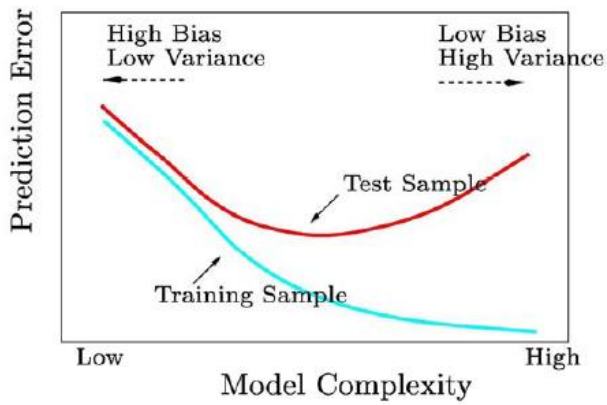
Linear Fit




9th order polynomial



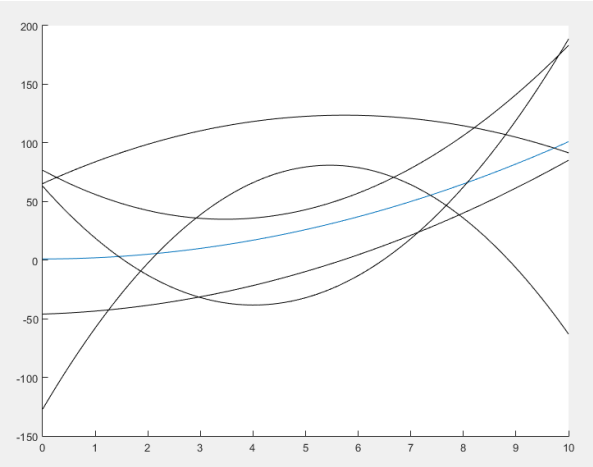
# Bias-Variance Dichotomy

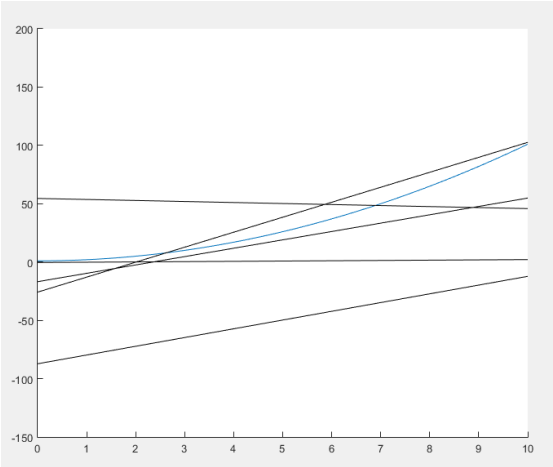


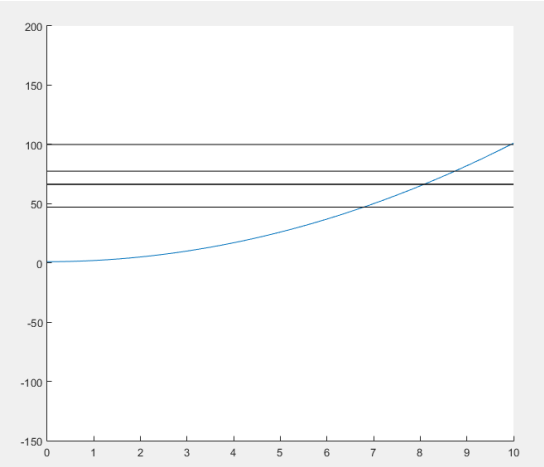

Bias-Variance Dichotomy from Hastie, Tibshirani, and Freidman (2001)

# Illustration 3

- Trying to fit a quadratic with 3 points:

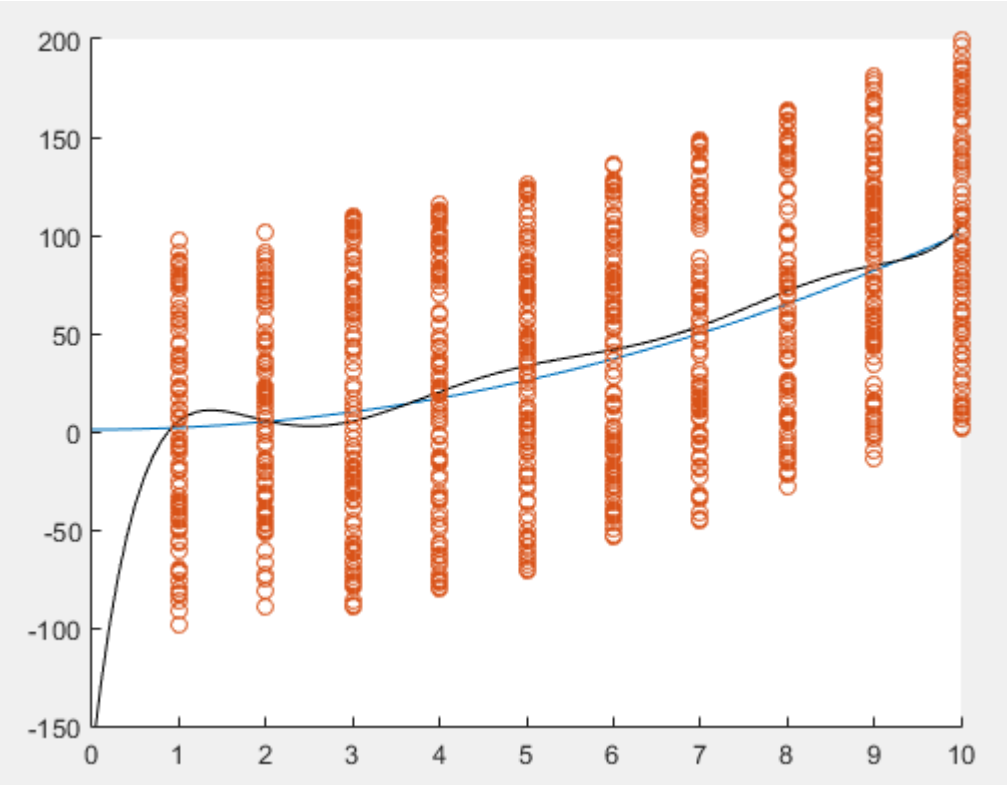


# Illustration 4

- What about fitting the quadratic with a ninth order polynomial (but with lots of points)



# Now that we also know K-NN

- How does the Bias-Variance dichotomy work for K-NN
- So what is Low-Bias, High Variance and what is the opposite?
- Think of K-NN where the input was Month-Date and the output was rainfall.

# Supervised Learning: Concepts

- Data Modelling versus Algorithmic modelling
- Bias-Variance Dichotomy
- Regularization and coefficient shrinkage
- Model Validation
- Dimensionality reduction

# Concept

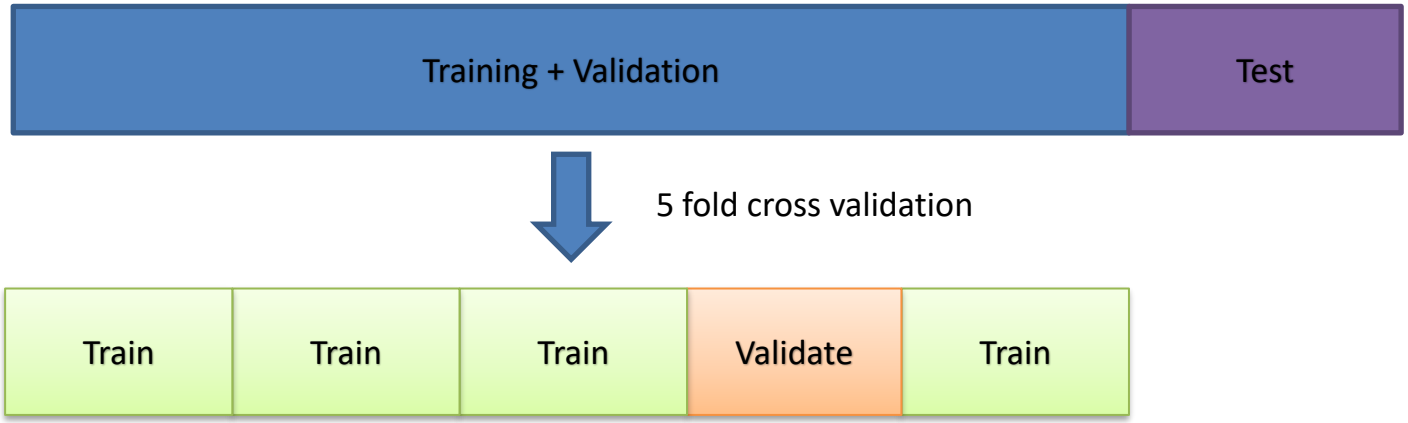
- Three separate objectives: Model training, Model selection, Model assessment
- Data break up for Training, Validation and Test

Or

# Break up

- Data for training. What happens in this phase.
- Validation phase: Data for model selection
  - Selecting between multiple methods
  - Fine tuning parameters in the algorithm. Affecting model complexity and regularization
  - Choosing the correct number of input variables and the correct input variables.
    - Affects model complexity.
    - Going beyond blanket approaches to penalize more inputs (Adj $R^2$, AIC, BIC, etc.)
- Test Phase: Data for model assessment

# Crossvalidation




- LOOCV (Leave One Out Cross Validation)

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# Regularization Techniques

- Variable selection: Penalty for using more variables
- Going beyond variable selection, what about variable shrinkage (Penalty for large opposing coefficients)
  - Multicollinearity and the potential for many forms of a regression equation
  - Y = 4A-2B or Y = 10A-8B. Actually prediction can also be a problem
- Ridge Regression/Lasso regression
- $\hat{\beta}^{ridge} = \min \sum_{i=1}^{n} (y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j)^2$

Subject to $\sum_{j=1}^{p} \beta_j^2 \leq s$

3 common approaches:

| Model | Penalty type |
|---|---|
| Ridge | $\|\beta\|_2$ |
| Lasso | $\|\beta\|_1$ |
| AIC/BIC/Adj $R^2$ | $\|\beta\|_0$ |

- $\hat{\beta}^{ridge} = \min\{\sum_{i=1}^{n} (y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j)^2 + \lambda \sum_{j=1}^{p} \beta_j^2\}$

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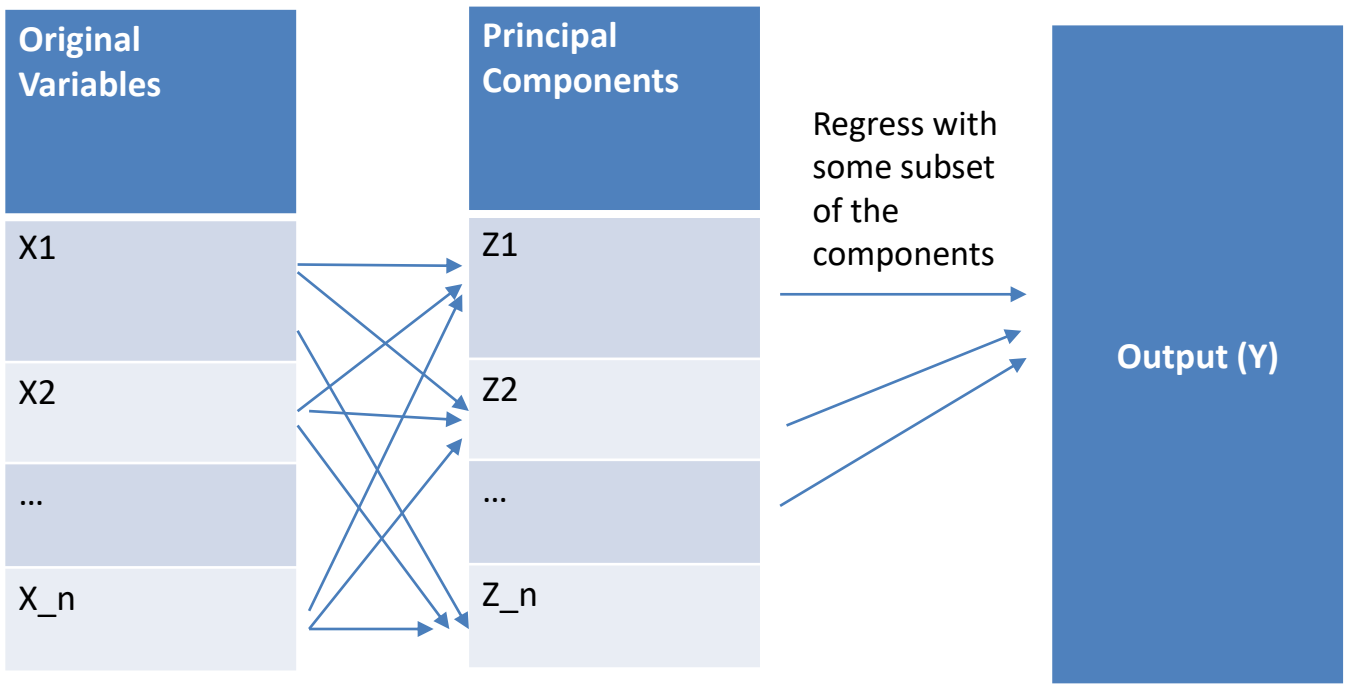
# Principal Component Analysis (PCA) and Regression (PCR)

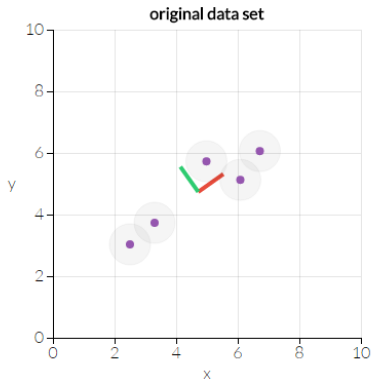
- We have already discussed how two highly correlated input variables can lead to problems of very large, unstable co-efficients.
- We corrected the problem of overfitting through Ridge and Lasso regression.
- Another approach is to transform the inputs
- We use PCA. How does PCA work?
- http://setosa.io/ev/principal-component-analysis/ (see backup slide)
- One example of where it can be useful? Imagine there is measurement noise of the input in addition uncorrelated noise on the output.

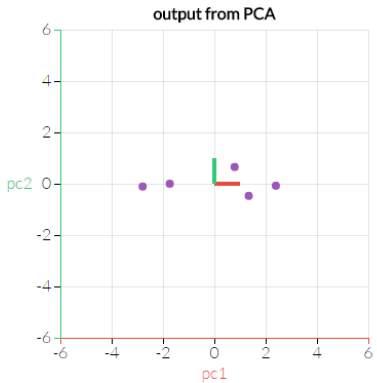
# PCR mapping

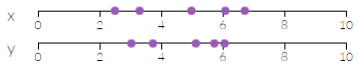



- What is Partial Least Squares?

# PCA - Backup




PCA is useful for eliminating dimensions. Below, we've plotted the data along a pair of lines: one composed of the x-values and another of the y-values.

If we're going to only see the data along one dimension, though, it might be better to make that dimension the principal component with most variation. We don't lose much by dropping PC2 since it contributes the least to the variation in the data set.